ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · B3 · CHANGING LANDSCAPES

Erosion words and solutions

★☆☆
Grade 4

Learning goals

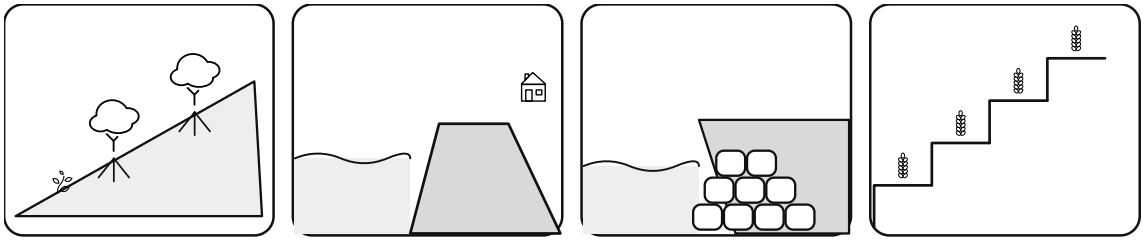
- Use erosion vocabulary.
- Name ways to slow erosion.

1 Match the word to its meaning. [4]

Draw a line to match each pair, or write the letter.

weathering ●	→ A	● **A** breaking rocks into small pieces
erosion ●	→ C	● **B** dropping mud and sand in a new place
deposition ●	→ B	● **C** carrying away soil and rock
landslide ●	→ D	● **D** soil and rocks sliding down a hill

2 Name each solution. [4]



A B C D

A	B	C	D
plant trees and grass	embankment	geo-bags	terraces

3 Sort the causes. [6]

SORT THESE

a river bank falling in **waves hitting a cliff** **sand blown onto a field**
sand dunes moving on a beach **rain washing soil off a hill**
dust storms in dry land

Water erosion	Wind erosion
waves hitting a cliff	sand blown onto a field
a river bank falling in	dust storms in dry land
rain washing soil off a hill	sand dunes moving on a beach

4 True or false? [3]

Statement	True	False	If false, correct it
1. Plant roots help hold soil in place.	✔		
2. Rivers only take land away; they never build new land.		✔	Rivers drop mud and sand that builds new land, such as chars.
3. A system is a group of parts that work together.	✔		

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · B3 · CHANGING LANDSCAPES

Choosing the best solution

★★☆
Grade 4

Learning goals

- Compare solutions.
- Explain using evidence.

1 **Why is erosion worse on a hill where all the trees have been cut down?** [2]

With no roots to hold the soil and no leaves to slow the rain, rain washes the soil straight down the hill, causing erosion and landslides.

2 **A village is beside a fast river that is eating away its bank. Compare two solutions and choose the best one. Explain why.** [4]

	Solution 1: _____	**Solution 2: _____**
good points	geo-bags: quick, cheaper	trees: cheap, natural
problems	can be washed away	too slow for a fast river now

e.g. geo-bags now, because they work quickly; then plant grass and trees behind them for the future.

3 **A coastal defence system has mangroves, an embankment and sluice gates. What would happen if the sluice gates were left open during a storm surge?** [2]

Sea water would flow through the gates into the fields behind the embankment, flooding them and making the soil salty.

4 **Draw a river bend. Show where erosion happens and where deposition happens.** [2]

River bend

Erosion on the outside of the bend (fast water); deposition (sand bar) on the inside (slow water).

ANSWER KEY

INVESTIGATE · B3 · CHANGING LANDSCAPES

Does a windbreak stop sand blowing away?

★★☆ Grade 4

Learning goals

- Carry out a fair test.
- Measure mass.
- Evaluate a solution.

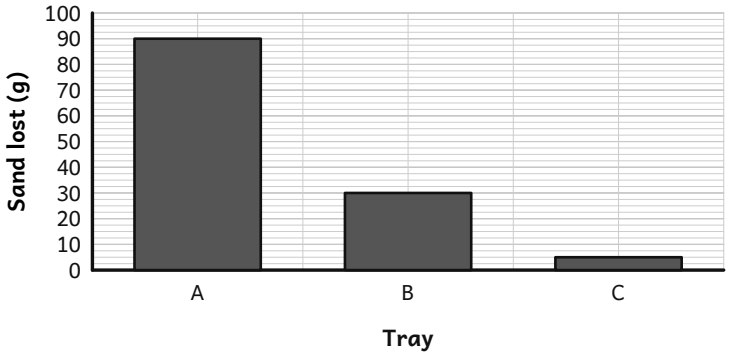
Class 4B put 200 g of dry sand in each of three trays. They blew air across each tray with a fan for 1 minute, then weighed the sand that was left. Tray A had nothing, tray B had a row of small sticks (a model windbreak), and tray C had a row of sticks and the sand was damp.

1 **Complete the plan.** [3]

Change	what protects the sand (nothing, sticks, sticks and damp)
Measure	mass of sand left (g)
Keep the same	amount of sand, fan speed, distance from the fan, time

2 **Complete the table and draw a bar chart of the sand lost.** [4]

Tray	Sand at start (g)	Sand left (g)	Sand lost (g)
A: nothing	200	110	90
B: windbreak	200	170	30
C: windbreak + damp	200	195	5



3 **Write a conclusion.** [2]

A windbreak greatly reduces wind erosion, and damp sand blows away even less.

4 **How could farmers use these results?** [1]

Plant rows of trees around fields and keep soil covered or damp.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · B3 · CHANGING LANDSCAPES

Living with a moving river

★★★
Grade 4

Learning goals

- Understand river erosion in Bangladesh.
- Use data about land loss.
- Compare solutions from different viewpoints.

Science · Geography · Maths · Economics · Citizenship

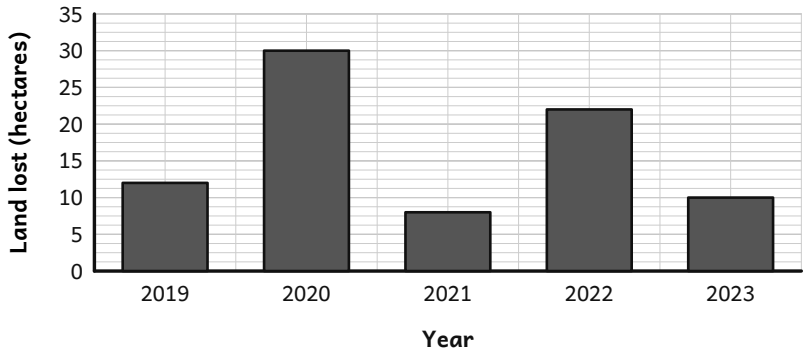
The **Jamuna** and **Padma** are among the biggest rivers in the world. Each monsoon they carry enormous amounts of water and sand. Their banks can collapse suddenly, and thousands of families lose their homes and fields to the river every year. People who lose their land often move to new chars or to the cities.

Engineers protect important places, such as towns and the Padma Bridge, with **concrete blocks**, **geo-bags** and walls along the banks. This is called **river training**. But it is expensive, and protecting one place can make the river push harder somewhere else.

Based on reports about riverbank erosion in Bangladesh. Data below are example values.

1 Land lost to the river at one village (example data). [4]

Year	2019	2020	2021	2022	2023
Land lost (hectares)	12	30	8	22	10



a) Draw the bar chart.

b) What is the total land lost in five years?

12 + 30 + 8 + 22 + 10 = 82 hectares

c) The worst years had very big monsoon floods. Why does this make sense?

More fast-moving water has more energy to wear away the banks.

2 **What might each person want?** [3]

Person	View
a farmer on the bank	Protect my village now with geo-bags; I will lose everything.
an engineer	Concrete works but is costly; we must check it does not push the river onto other villages.
a family on a new char	The river gave us new land, but floods can wash it away again.

3 **If you were in charge of the money, how would you protect the village? Explain your plan.** [2]

e.g. geo-bags at the worst places now, plant grass and trees on the bank, move the school away from the edge, and an early warning when the bank is cracking.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · B3 · CHANGING LANDSCAPES

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

T	I	M	K	L	W	I	N	D	G
V	G	H	T	G	I	L	D	N	T
R	N	K	E	K	A	B	K	A	L
I	B	C	H	A	R	O	O	T	S
V	S	C	E	R	O	S	I	O	N
E	Y	A	R	S	O	I	L	O	O
R	W	U	N	A	M	O	L	B	I
V	P	G	O	D	D	H	L	P	I
F	T	S	Y	S	T	E	M	W	E
Y	U	U	B	B	A	N	K	J	J

bank **char** **erosion** **river** **roots** **sand** **soil** **system** **wind**

2 Which is the best way to stop soil washing off a bare hill? [1]

- ○ cut down more trees
- ● **plant trees and grass**
- ○ build a road
- ○ burn the grass

3 Where does a river drop most of its mud? [1]

- ○ where it flows fastest
- ● **where it slows down**
- ○ at the top of a waterfall
- ○ on the outside of a bend

4 **Draw a coastal defence system with three parts. Label each part and its job.** [3]

Coastal defence system

mangroves (slow waves) → embankment (holds back sea) → sluice gate (lets rain water out, keeps sea out).

5 **Challenge. Why might protecting one part of a river bank make erosion worse further down the river?** [2]

The water cannot wear away the protected bank, so it is pushed with more force against the next unprotected bank, eroding it faster.